

1. Assignment: Ticket Booking System for Movies

MongoDB Database Schema

1. Collections:

- **Users**
 - **Movies**
 - **Theaters**
 - **Bookings**
 - **Roles [User and Admin]**
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Required APIs

User APIs

1. User Signup with Email Account Verification

- Allow users to sign up by providing their email, password, and other necessary details.
- Send an email verification link to the user upon signup.

2. User Login

- Allow users to log in using their email and password.

3. Get User Profile

- Fetch and display the user's profile information.

4. Edit User Profile (with Profile Picture)

- Update user profile details, including uploading or editing the profile picture.
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Movie Management APIs (Admin only)

1. Add Movie

- Add movie details including:
 - Movie name
 - Genre
 - Language
 - Duration
 - Cast
 - Director
 - Release date

2. Add Theater

- Add a new theater with details including:
 - Theater name
 - Location
 - Number of screens

3. Assign Movies to Theaters

- Assign specific movies to available screens in a theater.
- Include:
 - Theater ID
 - Screen number
 - Movie ID
 - Show timings

4. List of Movies

- Fetch a list of all movies available in the database.
- Include:
 - Total number of theaters playing that movie
 - Theaters and show timings

5. Edit Movie Details

- Update details for an existing movie.

6. Delete Movie Details

- Remove a movie from the database.
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Booking APIs

1. List of Theaters for a Movie

- Given a movie ID, fetch the list of theaters playing the movie along with show timings.

2. Book Tickets for a Movie

- Allow users to book tickets for a movie by providing:
 - Movie ID
 - User ID
 - Theater ID
 - Show timing
 - Number of tickets
- Update the availability of seats for the selected show.

3. Cancel Booking

- Allow users to cancel a booking by providing:
 - Booking ID
- Update the availability of seats for the corresponding show.

4. View Booking History

- Fetch and display the user's booking history.
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Reports APIs

1. List of Movies with Total Bookings

- Fetch a list of all movies along with the total number of tickets booked for each movie.

2. List of Bookings by Theater

- Fetch bookings grouped by theater, including:
 - Movie name
 - Total tickets booked for each show

3. Send Booking Summary to User Email

- Generate a summary of all bookings made by a user and send it to their email in table format.

2. Assignment: Institute Management System

MongoDB Database Schema

1. Collections:

- **Users** (Students, Teachers, Admins)
 - **Courses**
 - **Batches**
 - **Enrollments**
 - **Attendance**
 - **Exams**
 - **Roles [User and Admin]**
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Required APIs

User Management APIs

1. User Signup with Email Verification

- Allow users (students, teachers) to sign up by providing:
 - Name
 - Email
 - Password
 - Role (Student/Teacher).
- Send an email verification link to the user upon signup.

2. User Login

- Allow users to log in using their email and password.

3. Get User Profile

- Fetch and display the user profile details (students/teachers).

4. Edit User Profile

- Update user profile details, including:
 - Profile picture
 - Contact information

5. List of Users (Admin Only)

- Fetch a list of all users in the system based on roles (Students, Teachers).
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Course Management APIs

1. Add Course (Admin Only)

- Add details for a new course including:
 - Course name
 - Description

- Duration
 - Fees
2. **Edit Course Details (Admin Only)**
 - Update course information.
 3. **Delete Course (Admin Only)**
 - Remove a course from the system.
 4. **List of Courses**
 - Fetch all available courses along with:
 - Total batches for the course
 - Total enrolled students
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Batch Management APIs

1. **Add Batch (Admin/Teacher)**
 - Create a batch for a specific course with:
 - Batch name
 - Start date
 - End date
 - Assigned teacher
 2. **Assign Students to Batch (Admin Only)**
 - Enroll students into a specific batch.
 3. **List of Batches**
 - Fetch all batches for a course, including:
 - Batch name
 - Total students
 - Assigned teacher
 4. **Update Batch Details (Admin/Teacher)**
 - Modify batch information such as schedule or assigned teacher.
 5. **Delete Batch (Admin Only)**
 - Remove a batch from the system.
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Enrollment and Attendance APIs

1. **Enroll Student into a Course**
 - Allow students to enroll in a course by providing:
 - Student ID
 - Course ID
2. **Mark Attendance (Teacher)**
 - Input attendance details:
 - Batch ID

- Date
- List of present and absent students

3. **View Attendance (Student/Teacher)**

- Fetch attendance records by:
 - Student (for a specific course or batch).
 - Batch (view attendance percentage for all students).
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Exam Management APIs

1. **Create Exam (Admin/Teacher)**

- Add an exam for a batch including:
 - Exam name
 - Date
 - Duration
 - Total marks

2. **Assign Marks to Students (Teacher)**

- Input marks for students for a specific exam.

3. **Fetch Exam Results (Student/Teacher)**

- View results by:
 - Student (individual performance).
 - Batch (all student results).

4. **Update Exam Details (Teacher)**

- Modify exam information such as date or marks distribution.
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Reports APIs

1. **List of Courses with Total Enrollments**

- Fetch all courses along with the total number of enrolled students.

2. **Batch Performance Report (Admin/Teacher)**

- Fetch a report for a specific batch including:
 - Attendance percentage for all students.
 - Average exam performance for the batch.

3. **Student Performance Report**

- Fetch a student's overall performance, including:
 - Attendance percentage.
 - Average marks for all exams.

4. **Send Student Report to Email**

- Generate a detailed performance report for a student and email it in table format.

3. Assignment: *Daily Task To-Do Planner*

MongoDB Database Schema

1. Collections:

- **Users**
 - **Tasks**
 - **Categories**
 - **Labels**
 - **Reminders**
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Required APIs

User Management APIs

1. User Signup with Email Account Verification

- Allow users to sign up by providing:
 - Name
 - Email
 - Password
- Send an email verification link to the user upon signup.

2. User Login

- Allow users to log in using their email and password.

3. Get User Profile

- Fetch and display the user's profile details.

4. Edit User Profile

- Update user profile details, including:
 - Name
 - Email
 - Profile picture
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Task Management APIs

1. Add Task

- Create a new task with details:
 - Task title
 - Description
 - Priority (Low, Medium, High)
 - Due date and time
 - Category ID (optional)

- Labels (optional)

2. Edit Task

- Update task details such as title, description, priority, due date, category, or labels.

3. Delete Task

- Delete a task by its ID.

4. Mark Task as Completed

- Update the status of a task to "Completed."

5. List Tasks

- Fetch all tasks for a user with optional filters:
 - By due date (e.g., tasks for today, tomorrow, this week).
 - By status (Pending, Completed).
 - By category or labels.

6. Reorder Tasks

- Change the order of tasks in the list for prioritization.
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Category and Label APIs

1. Add Category

- Create a new category for grouping tasks.
 - E.g., Work, Personal, Fitness

2. Edit Category

- Update category details such as name or description.

3. Delete Category

- Remove a category.

4. Add Label

- Create a custom label for tagging tasks.
 - E.g., "Urgent," "Meeting," "Chores"

5. List Categories and Labels

- Fetch all categories and labels for the user.
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Reminder Management APIs

1. Set Reminder for Task

- Add a reminder for a specific task.
 - Reminder types:
 - Notification at a specific time
 - Repeating reminders (e.g., daily, weekly)

2. Edit Reminder

- Update the reminder settings for a task.

3. Delete Reminder

- Remove a reminder for a task.
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Reports and Insights APIs

1. Task Summary for a Day or Week

- Fetch a summary of tasks for a specific day or week, including:
 - Total tasks
 - Completed tasks
 - Pending tasks

2. Task Statistics

- Generate insights on task completion rates:
 - Percentage of completed tasks over time.
 - Average time to complete tasks.

3. Send Daily/Weekly Task Summary to Email

- Automatically email the user a summary of tasks, including:
 - Overdue tasks
 - Upcoming tasks
 - Completed tasks